# Nisong Monyimba

1515 S Extension Rd, Mesa, AZ 85210 | nmonyimb@asu.edu | 602-545-6789 |
github.com/NisongMonyimba/AppliedMathThesis

Applied mathematician with a background in biomedical engineering whose research focuses on stochastic analysis, mean-field games, stochastic control, and scientific computing. Current work develops analytical and computational methods for common-noise mean-field games, combining stochastic differential equations, optimal transport, numerical analysis, and reproducible research software. Seeking to pursue this work through doctoral study in applied mathematics.

## Research Interests

Stochastic Analysis • Mean-Field Games • Numerical Analysis • Scientific Computing • Stochastic Control • Optimal Transport • Partial Differential Equations

## Education

**Biomedical Engineering, M.Sc. — Arizona State University** Graduated May 2022

*Ira A. Fulton Schools of Engineering; MasterCard Foundation Scholar; Dean's List: Fall 2020, Spring 2021*

**Biomedical Engineering, B.Sc. — KNUST** Graduated May 2021

*Kwame Nkrumah University of Science & Technology; MasterCard Foundation Scholar; First Class Honours*

**Undergraduate quantitative coursework:** Calculus, Differential Equations, Numerical Methods, Probability and Statistics, Signals and Systems.

## Independent Mathematical Preparation

**Graduate-Level Self-Study** 2025–Present

*Undertaken independently of formal coursework, in connection with the research program below*

- Real Analysis, Functional Analysis, Measure-Theoretic Probability, Stochastic Calculus, Mean-Field Games, McKean–Vlasov SDEs, Optimal Transport, and Numerical PDEs

## Research

**Independent Research Program in Applied Mathematics** 2025–Present

*Mean-field games • stochastic control • Wasserstein geometry • optimal transport • FBSDEs • numerical PDEs • Lean 4 formal verification*

Current research develops mathematical and computational methods for common-noise mean-field games, combining stochastic analysis, numerical analysis, scientific computing, and reproducible research software. Current outputs: a theoretical preprint, a numerical preprint, a software repository, and an independent research monograph (below).

### Selected Research

*Quantitative Contraction Estimates for Common-Noise Mean-Field Games* (2026). Establishes a local contraction estimate, on an explicit time horizon, for the fixed-point map characterising equilibria of mean-field games with common noise, extended to arbitrary horizons via a continuation argument.
PDF | GitHub Repository

*Numerical Approximation of Common-Noise Mean-Field Games via Particle Methods and Finite Differences* (2026). Combined Euler–Maruyama, particle-method, and finite-difference scheme for common-noise mean-field game equilibria, with every reported convergence rate the direct, reproducible output of executed code.
PDF | GitHub Repository

### Independent Research Monograph

*Quantitative Convergence Analysis of Stochastic Maximum Principles for Mean-Field Games with Common Noise* (2026). A book-length treatment developed independently following the 2022 M.Sc., covering well-posedness, particle methods, finite-difference schemes, and partial Lean 4 formalisation.
PDF | Full repository

### Current Research Directions

- Nonlinear (non-LQ) common-noise mean-field games
- Numerical stochastic control and common-noise master equations
- Numerical methods for high-dimensional FBSDEs and PDEs
- Formal verification of stochastic analysis results in Lean 4

## Research Software

**AppliedMathThesis (Python, Lean 4)**

- Research software implementing numerical methods for common-noise mean-field games
- Euler–Maruyama and Milstein SDE solvers, including mean-field/common-noise variants
- Particle-method implementation with conditional propagation-of-chaos verification
- Finite-difference Fokker–Planck solver
- Wasserstein-distance computations for convergence verification
- Automated numerical benchmarks (convergence rates, runtime scaling) and reproducible computational experiments
- Lean 4 formalisation of core estimates (Gronwall, contraction bound, propagation of chaos)
- Version-controlled, continuously integrated (GitHub Actions), automatically tested (pytest)
- github.com/NisongMonyimba/AppliedMathThesis

## Research Experience

**Research Assistant, Weaver Lab, Arizona State University** 2021–2022

- Analyzed trophoblast proliferation dynamics by quantifying cell counts over time and fitting growth curves.
- Used Fiji to extract quantitative metrics from laboratory images, including colony expansion area and fluorescence intensity, supporting rigorous modeling of cellular growth patterns.
- Quantified cell proliferation rates from fluorescence microscopy data using Fiji and GraphPad Prism; developed statistical plots and regression analyses to characterize proliferation and viability.
- Developed Python-based probabilistic simulations and Monte Carlo analyses to evaluate variability in material properties and stochastic loading, improving predictive accuracy.

**Research Assistant, Dr. Pizziconi's Lab, Arizona State University** 2022–2023

- Designed and optimized bone fixation screws using Halpin–Tsai micromechanics models to predict composite behavior under mechanical load.
- Created and refined 3D CAD models in SolidWorks, ensuring dimensional accuracy, manufacturability, and integration with experimental validation workflows.
- Trained and mentored lab members in modeling, simulation, and experimental validation, connecting theoretical mathematics with engineering practice.
- Collaborated with interdisciplinary teams to iterate designs from simulation results; documented and presented research outcomes supporting publication efforts and potential clinical translation.

**Research Assistant, Medical Device Regulation (MDR) Project, KNUST** 2020–2021

- Analyzed regulatory frameworks governing medical devices across 20 African countries, identifying disparities in approval processes, safety standards, and market accessibility.
- Used publicly available regulatory data to categorize countries by regulatory maturity, from fully established authorities to emerging or informal frameworks.

**Med Tech R&D Engineer, Mayo Clinic & Arizona State University** 2022–2023

- Designed bone fixation screws using Halpin–Tsai equations and Mori–Tanaka models.
- Ran nonlinear FEA simulations to analyze fatigue, stress distribution, and deformation behavior.

## Teaching Experience

*Teaching philosophy:* I aim to teach mathematics by combining rigorous theory with computational examples and reproducible scientific programming.

**Teaching Interests**

*Based on completed undergraduate coursework, self-directed graduate-level study, and the teaching and tutoring experience below*

- Calculus I & II, Linear Algebra, Differential Equations, Numerical Analysis, Probability & Statistics, Engineering Mathematics, Scientific Computing (Python)

**Teaching Assistant, KNUST** 2019

- Tutored biomedical engineering undergraduates in calculus, electrical circuits, and biostatistics; assisted with lab supervision, grading, and practical demonstrations.

**Science & Math Teacher, NSMQ Team, Kwanyako Senior High School** 2016–2018

- Prepared and led students for regional and national science and math competitions, emphasizing teamwork, critical thinking, and problem-solving.

**Math Teacher, Witness Academy** 2016–2017

- Taught core mathematics and physics; prepared junior-high students for final exams (BECE).

## Master's Applied Project & Undergraduate Capstone Project

**Master's Applied Project: Trophoblast-on-Chip Microfluidic Device** 2022

- Investigated trophoblast growth to quantify proliferation dynamics (cell counts over time, growth curves, proliferation index) using Fiji to extract quantitative growth metrics from laboratory cell images.
- Performed statistical analysis including growth-curve comparison across conditions (two-way ANOVA), doubling-time comparison between treatments (unpaired t-test), and proliferation-marker quantification (Kruskal–Wallis).
- Developed microfluidic channel designs in SolidWorks, simulated laminar flow and shear profiles in COMSOL Multiphysics, and validated models against analytical equations.

**Capstone Project: Novel ECMO Cannula Design, Arizona State University** 2021

- Designed and simulated ECMO cannula flow in COMSOL Multiphysics, optimizing shear stress and velocity distribution; built a 3D model in SolidWorks and simulated flow in ANSYS Fluent.
- Quantified flow uniformity and reduced clot-formation risk by optimizing inlet geometry and flow-rate parameters.
- Performed finite element analyses (FEA) in ANSYS and SolidWorks to assess performance and safety margins under physiological load conditions.

## Internship Experience

**REU Internship, Case Western Reserve University** 2020

- Studied biological design and product development principles; designed Arduino-based air-quality sensors and prototyped PCB systems using EAGLE CAD.
- Collaborated with interdisciplinary teams to apply quantitative modeling to life-science problems.

**BME Internship, Komfo Anokye Teaching Hospital, Kumasi, Ghana** 2019

- Supported medical equipment maintenance and calibration; performed voltage and current testing.
- Documented device performance data and proposed engineering solutions to minimize error variance; presented findings to senior biomedical engineers and clinicians.

**Engineering Internship, Vodafone (Telecom) Headquarters, Accra, Ghana**

- Designed and proposed a telemedicine smartphone application concept for patient monitoring.
- Assisted in vendor management and quality audits for customer-facing telecom hardware systems.

## Professional Experience

**Senior Manufacturing Engineer, Abbott Laboratories** 2024–2025

- Optimized medical device grinding precision by 31.7% through DOE, ANOVA, and SPC-driven process improvements.
- Applied Monte Carlo simulations, PCA, and variance component analysis to identify and mitigate process variability.

**Quality Engineer, Vastek Inc., San Diego, CA** 2023–2024

- Designed devices and implemented engineering design practices targeting efficiency improvements; worked with a cross-functional team of medical professionals and engineers.
- Conducted FMEA, Bayesian risk analysis, and fault-tree analysis to quantify cascading failure probabilities.

## Volunteering, Community Service & Projects

- **Group Homes Volunteer (Autism Support):** assisted residents with learning and recreational activities.
- **MasterCard Foundation Community Projects:** led sanitation, clean-water, and youth education projects.
- **Oforikrom Outreach Project:** organized public school clean-ups, donations, and science tutoring.
- **Vocational Training:** trained four youths in electrical wiring and biofill installation.
- **Personal Tutoring Initiative:** mentored underprivileged students in mathematics; three students gained high-school admission and two gained college admission, all first-generation in their families.
- Organized community cleaning (gutters, weeds, trash) in Ashaiman-Zenu to reduce contamination and malaria risk.

## Awards & Fellowships

- MasterCard Foundation Scholarship, Arizona State University and KNUST
- MORE Fellowship Award, Arizona State University
- Vodafone Telecommunications Design Competition, 1st Place
- 2nd Place, KNUST Engineering & Medical School Innovation Challenge
- Dean's List, Arizona State University

## Leadership & Mentorship

- Peer Mentor, MasterCard Foundation Scholars (KNUST & ASU)
- Community Project Facilitator, MasterCard Foundation Ghana
- Member, Young African Leaders Initiative (YALI)
- Club Executive, Biomedical Engineering Society (KNUST & ASU)

- Leader, NSMQ Science & Math Quiz Team, Kwanyako SHS

## Certifications

- Lean Six Sigma (Green Belt)
- Design of Experiments (Minitab & JMP)
- Global Leadership & Project Management (YALI / ASU Global Leadership Academy)

## Professional Memberships

Society of Manufacturing Engineers (SME) • National Society of Black Engineers (NSBE) • Biomedical Engineering Society (BMES) • ASU Global Leadership Academy • Toastmasters International • KNUST Peer Counsellors & MasterCard Peer Mentors • Ghana Engineering Students Association • TraTech

## Mathematical Software

Python (NumPy, SciPy, pandas, statsmodels) • Lean 4 • MATLAB • GNU Octave

## Scientific Computing

Finite-Difference Methods • Monte Carlo Simulation • Euler–Maruyama & Milstein Schemes • Particle Methods • Numerical Linear Algebra • Computational Complexity Analysis • Numerical Verification • Reproducible Research

## Formal Mathematics

Lean 4 • formalisation of selected results in stochastic analysis, including Gronwall's inequality, contraction bounds, and propagation-of-chaos lemmas. Repository includes an automated compilation check (GitHub Actions, `lake build`) against current Mathlib releases.

## Technical Skills

**Programming & Version Control:** Python, MATLAB, R, Git/GitHub, LaTeX (Overleaf).

**Engineering Simulation, CAD & FEA:** SolidWorks, ANSYS Fluent, COMSOL Multiphysics, Autodesk Fusion 360, SimScale, finite element methods.

**Statistical & Process Methods:** ANOVA, PCA, DOE, SPC, optimization, GraphPad Prism, Minitab, JMP.

**Other:** Fiji/ImageJ, BioRender, Figma, Inkscape, LabVIEW, LTSpice, Windchill, Zotero.

## Languages

English • Twi